TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

# Direct and Indirect (Stock Solution) Calculation from a Recipe

*The calculation-led foundation of all laboratory dyeing: converting % owf and g/L into weighings and pipette volumes that a bulk recipe can trust.*

TE-3106 · TEXTILE CHEMISTRY & COLORATION-II SESSIONAL · EXPERIMENT 1 · LEVEL 3 TERM 1

## 1. Aim and theory

**Aim:** to prepare accurate stock solutions and to calculate — directly and indirectly — every quantity needed for a laboratory dyeing, so that a 5 g lot reproduces a bulk recipe exactly. Two routes exist. **Direct calculation** weighs each dye and chemical for the bath. **Indirect calculation** prepares a **stock solution** of known strength and pipettes the required volume; it is the only accurate route when the dye mass is below about 0.05 g, where balance error dominates.

$$\text{Mass needed (g)} = (\%\ \text{owf} \times \text{fabric mass in g})/100 \quad | \quad \text{Stock volume (mL)} = \text{mass needed (mg)} \div \text{stock strength (mg/mL)}$$

Stock strengths are written % w/v: 1% = 1 g per 100 mL = 10 mg/mL; 2% = 20 mg/mL; 0.5% = 5 mg/mL. Bath chemicals quoted in g/L convert through bath volume: mass (g) = g/L × bath (L), with bath (L) = fabric (kg) × LR.

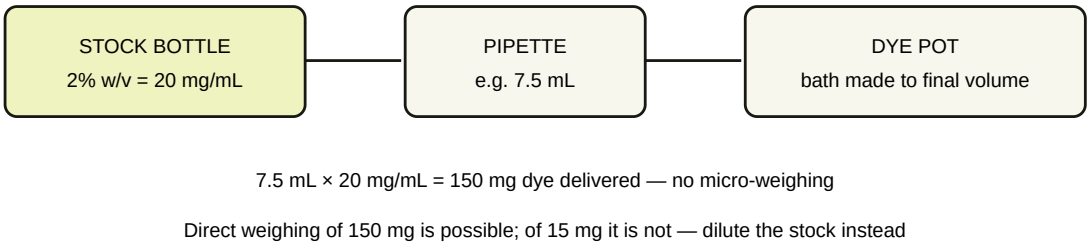



Fig. 1 — Indirect dispensing: stock solution and pipette replace micro-weighing.

## 2. Apparatus and materials

- Analytical balance (±0.001 g), volumetric flasks (100/250 mL), pipettes (5–25 mL), beakers, dye pots, laboratory dyeing machine.
- Dye (single strength-standardised lot), distilled water, fabric lots of 10.0 g scoured cotton.
- Stock preparation: paste the dye with a little cold water, dissolve in hot distilled water, cool, make to the mark, mix, label with dye name, CI name, strength, date. Stocks keep 1–2 weeks cool and dark; reactive dye stocks hydrolyse slowly and are made fresh weekly.

## 3. Procedure and worked calculations

**Job A — direct and indirect dye mass.** Lot 10.0 g, recipe 1.50% owf.

> **CALCULATION A**
>
> Dye mass = 1.50 × 10.0/100 = 0.150 g = 150 mg.
> From 2% stock (20 mg/mL): V = 150/20 = **7.5 mL**.
> From 1% stock (10 mg/mL): V = **15.0 mL**.
> From 0.5% stock (5 mg/mL): V = **30.0 mL** — already a large fraction of a 200 mL bath; choose stock strength so the pipetted volume sits between 2 and 20 mL.

**Job B — full bath build-up.** Same lot, LR 1:20 → bath 200 mL; salt 40 g/L; soda ash 15 g/L; wetting agent 1 g/L.

> **CALCULATION B**
>
> Salt = 40 × 0.200 = **8.00 g**. Soda ash = 15 × 0.200 = **3.00 g**. Wetting agent = 1 × 0.200 = **0.20 g** (dispense from a 10% stock: 2.0 mL).
> Bath assembly: pipette dye stock 7.5 mL + dissolve salts in ~150 mL water + wetting stock 2.0 mL, then **make up to 200 mL total**. Final-volume discipline keeps every g/L exactly as calculated.

**Job C — preparing the stock itself.** 2% w/v in a 250 mL flask: 2 g per 100 mL → **5.00 g** dye made to 250 mL. If the dye lot is certified 90% strength against the standard, weigh 5.00/0.90 = **5.56 g** so the stock matches standard-shade recipes.

## 4. Observation / record table (as recorded)

| ITEM | CALCULATED | DISPENSED | CHECK |
|---|---|---|---|
| Dye 1.50% owf | 150 mg | 7.5 mL of 2% stock | 150 mg ✓ |
| Salt 40 g/L | 8.00 g | 8.01 g weighed | +0.1% ✓ |

| | | | |
|---|---|---|---|
| Soda ash 15 g/L | 3.00 g | 3.00 g | ✓ |
| Bath volume | 200 mL | made to 200 mL mark | LR 1:20 ✓ |

## 5. Precautions

- Never return pipetted excess to the stock bottle — contamination and hydrolysis follow.
- Wet the pipette with the stock before measuring (rinse-drain); read the meniscus at eye level.
- Record stock age: a three-week-old reactive stock dyes measurably paler — hydrolysis is silent.
- Label every pot with the full recipe before heating; unlabelled pots are discarded, not guessed.

## 6. Practice (viva preparation)

1. Lot 5.0 g, dye 0.80% owf, 1% stock. Calculate the pipette volume.
2. LR 1:15, lot 10 g, alkali 10 g/L. Find bath volume and alkali mass.
3. Why is a 0.1% stock unsuitable for dispensing a 2% owf depth on a 10 g lot?
4. A stock is made from a dye lot of 85% strength without correction. What happens to every shade dyed from it?

### Answer key

1. Mass = 0.80 × 5.0/100 = 0.040 g = 40 mg; 1% = 10 mg/mL → 4.0 mL.
2. Bath = 10 g × 15 = 150 mL = 0.150 L; alkali = 10 × 0.150 = 1.50 g.
3. Dye needed = 200 mg; 0.1% stock = 1 mg/mL → 200 mL of stock — larger than the whole bath. Stock strength must match the dose scale (2–20 mL rule).
4. Every bath receives 15% less dye than calculated; all shades come up pale by a consistent factor — the signature of a strength error, corrected by weighing mass/0.85.

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